the Nansen LEGACY

Annual Meeting

25. - 27. October 2021

Modeled vs. natural spatial distributions: How far is reality from the Ecospace Barents Sea model?

Marcela C. Nascimento; Raul Primiciero and Torstein Pedersen

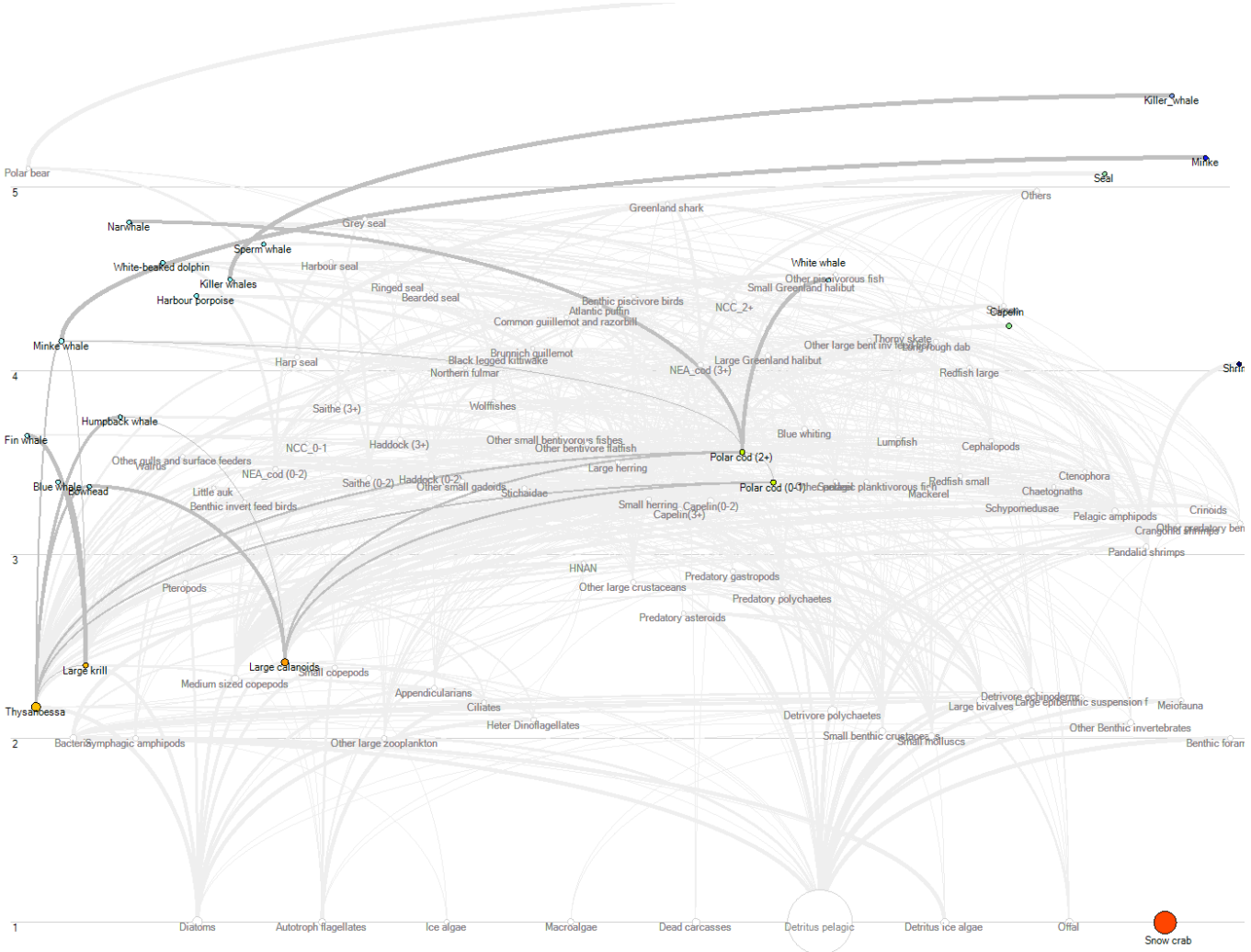
THE ARCTIC UNIVERSITY OF NORWAY • UIT •

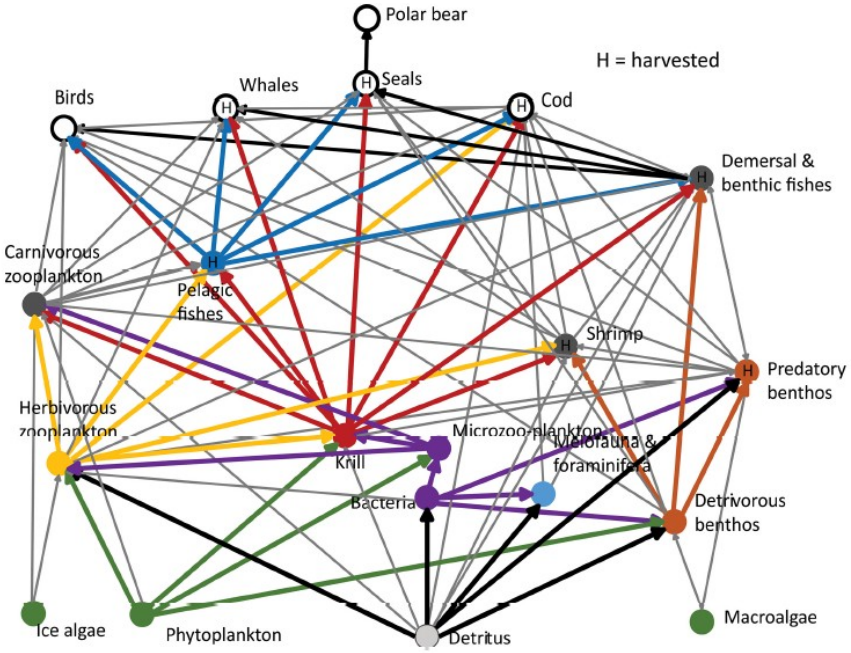
frontiers in Marine Science

ORIGINAL RESEARCH
published: 17 September 2021
doi: 10.3389/fmars.2021.732637

Warming of the Barents Sea Ecosystem During 1950–2013

Torstein Pedersen[1], Nina Mikkelsen[1,2], Ulf Lindstrøm[1,2], Paul E. Renaud[3,4], Marcela C. Nascimento[1], Marie-Anne Blanchet[5†], Ingrid H. Ellingsen[6], Lis L. Jørgensen[2] and Hugues Blanchet[3,7]*





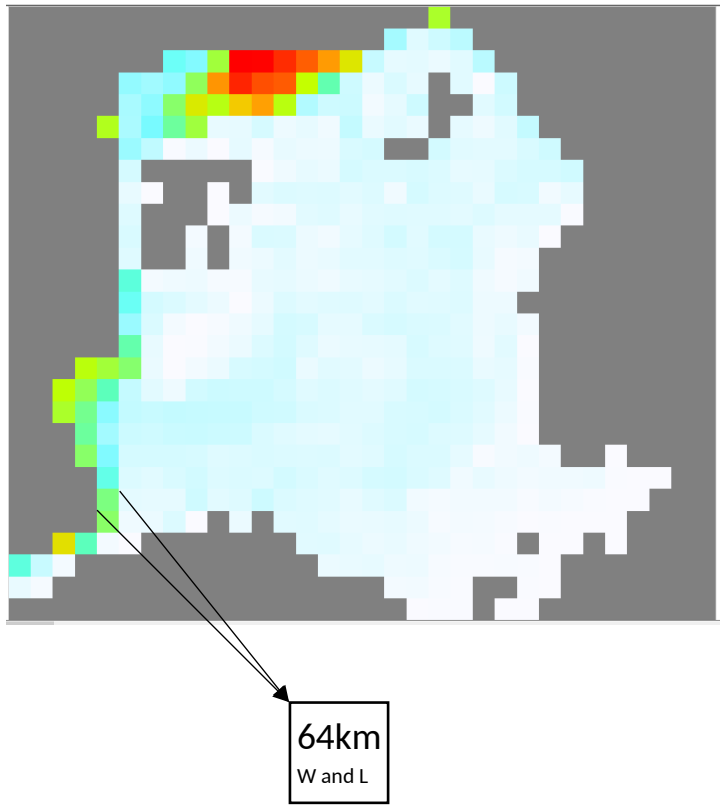
108 FG
ca 300 species

Trophic flow

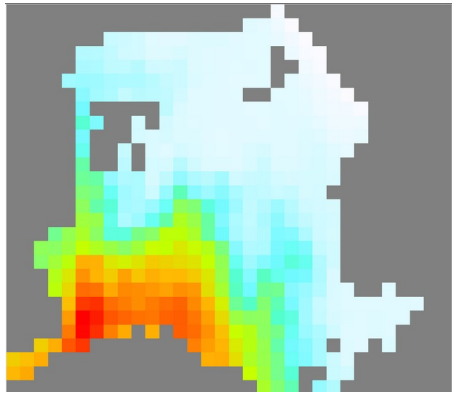
1998 **2013**

Cold **Warm**

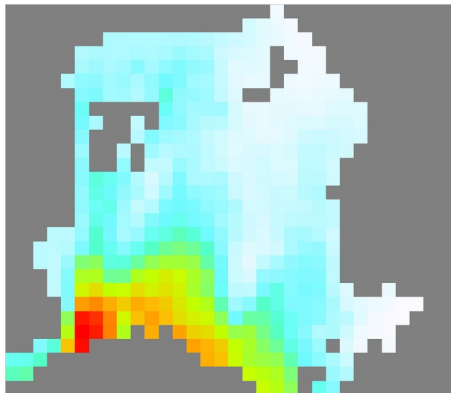
Depth



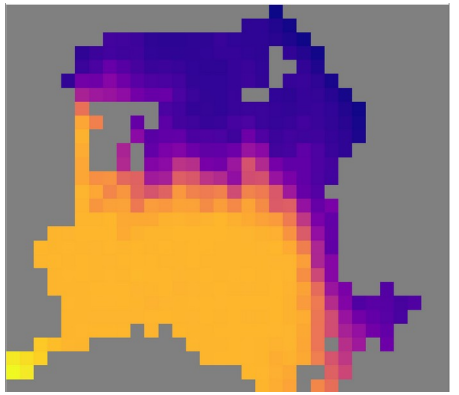
Surface T°
(1998)



Bottom T°
(1998)



Ice coverage
(1998)



Average 0.4°C

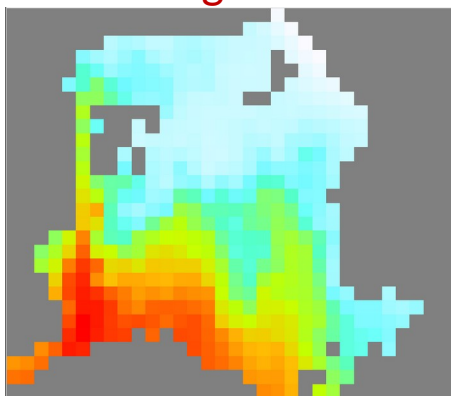
Average 0.3°C

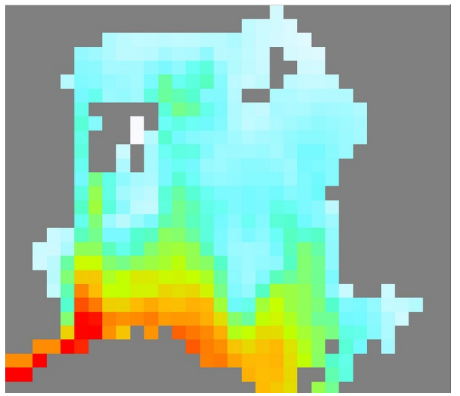
Kola (0-200) = 3,6

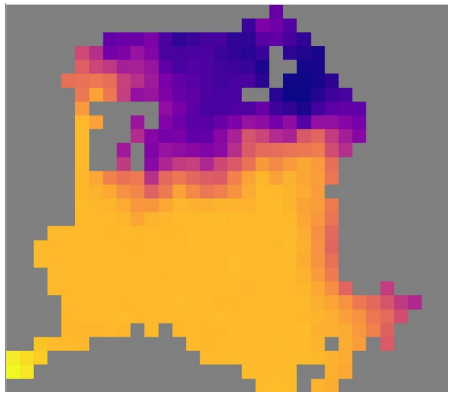
Kola (0-200) = 4,9

Average 1.3°C

Average 1.3°C







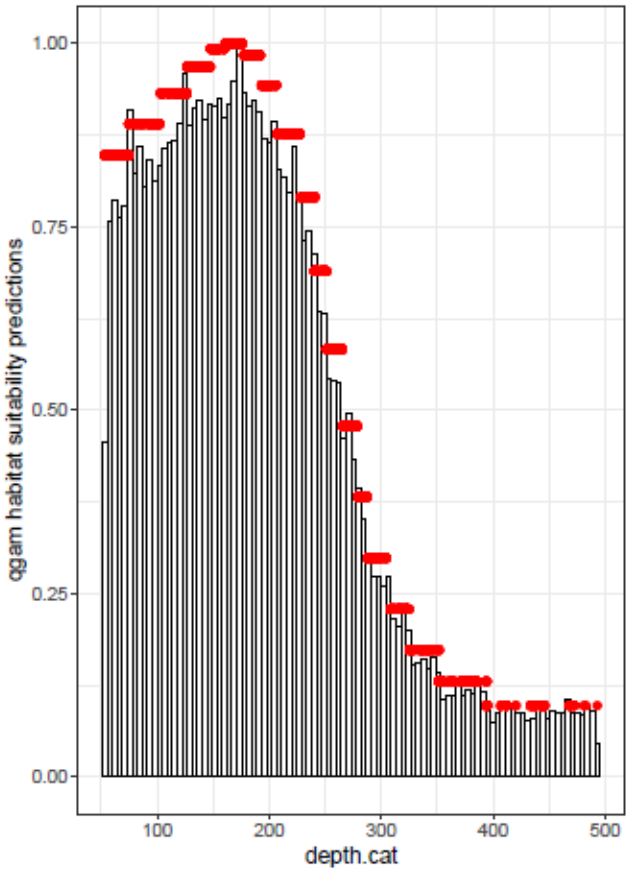
Surface T°
(2013)

Bottom T°
(2013)

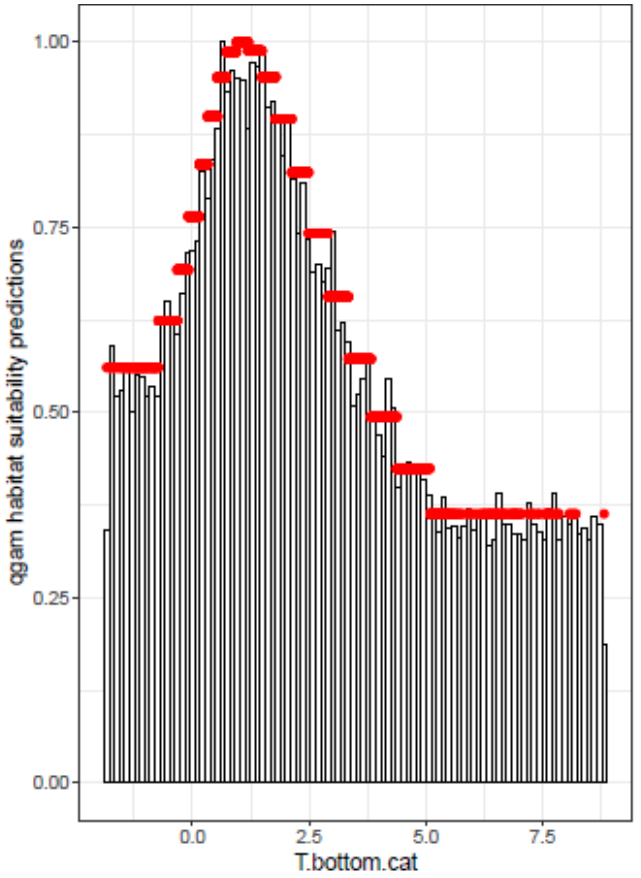
Ice coverage
(2013)

Environmental condition – Ecosystem survey
IMR/PINRO - SVIM archives

Habitat capacities



Depth



Bottom temperature

Cod



www.oceana.org

77 FG
1 or + environmental response

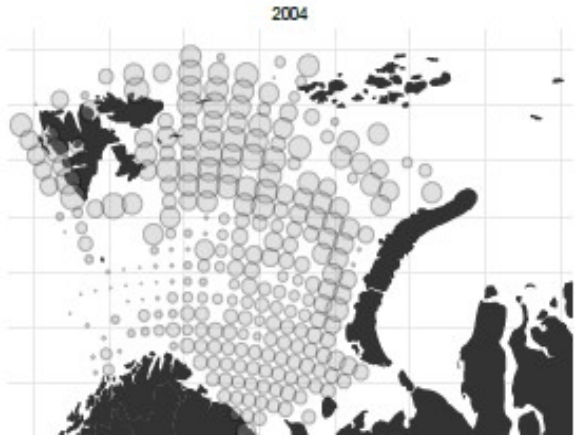
Functional groups with higher biomass
37 FG

Thank you Bérengère Husson and Lilia Guillet

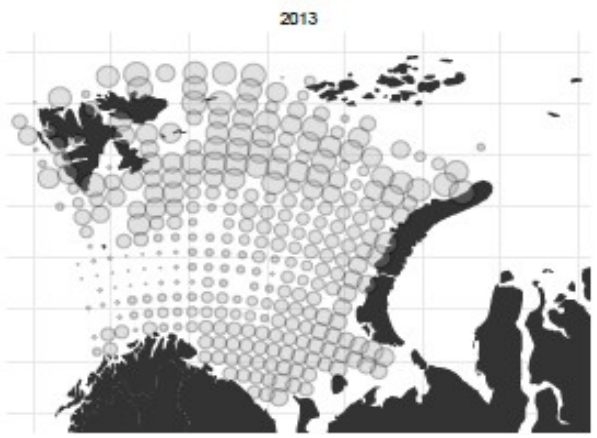
QGAM model – ecosystem survey/literature

Biomass distribution

2004 “less warm”



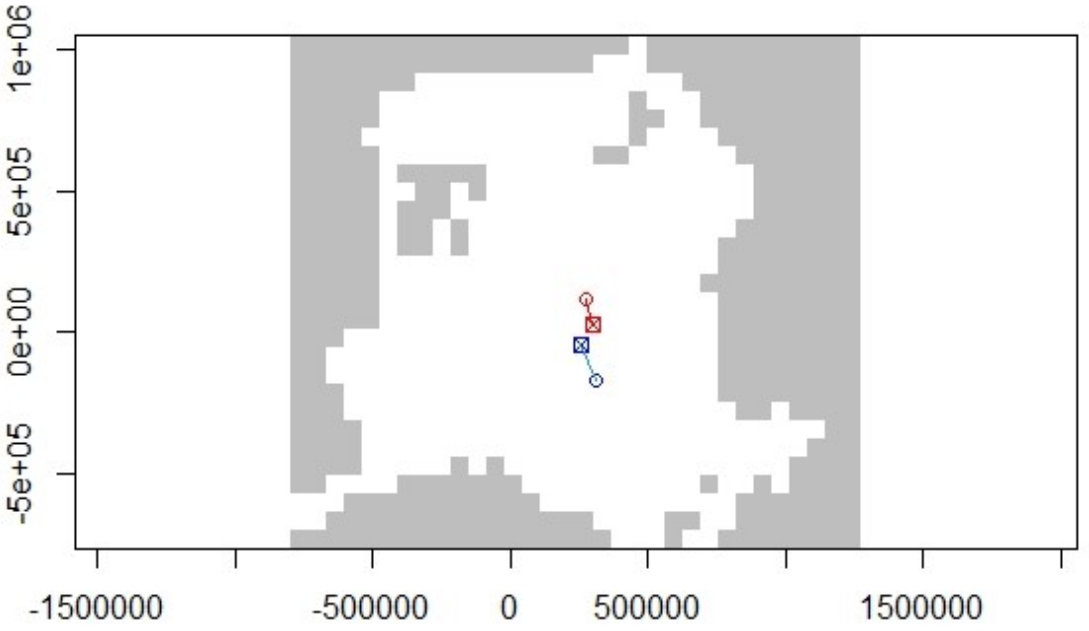
2013 warm



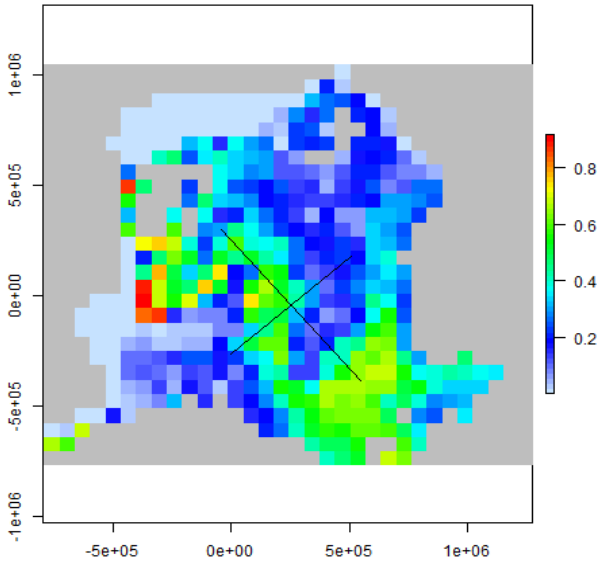
Cod



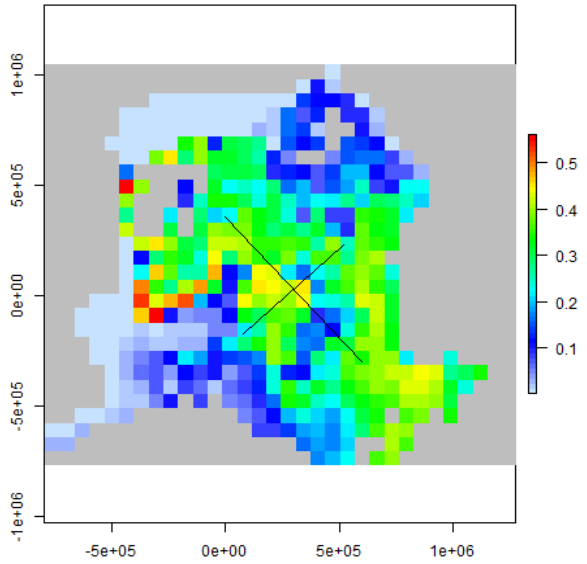
www.oceana.org



1998 cold

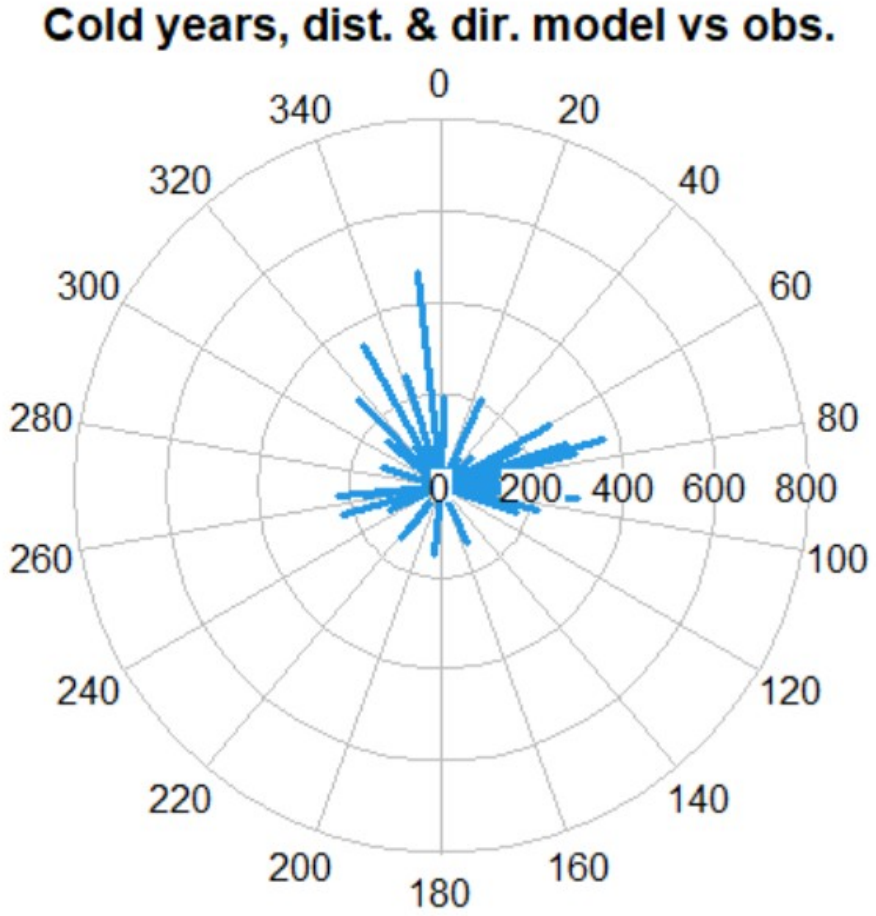


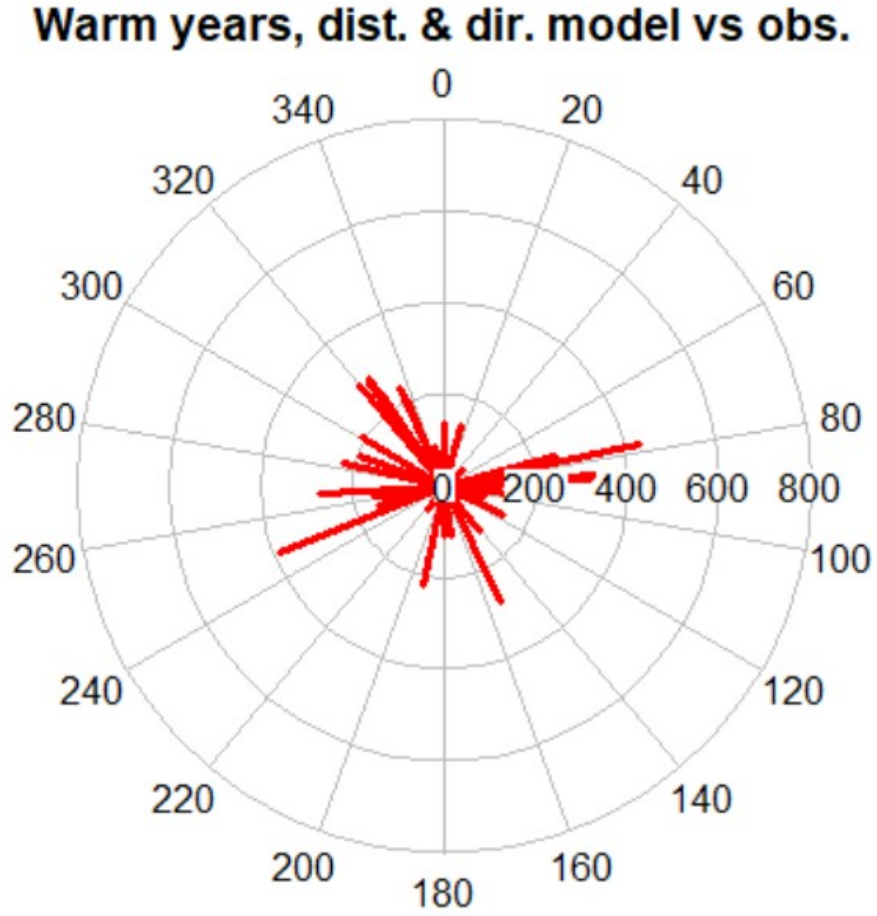
2013 warm



Center of gravity and *inertia*

Distance in Km between modeled and observed CG			Azimuth	
Species	cold	warm	cold	warm
NEA COD 3+	139.3	87.3	156	346





Average distance	191 km	194.4km
SD	94	98

Thank you